

A Web-Based Public Grievance Redressal System with Image and Geolocation Support

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Abstract--This paper presents Infra-Alert, a crowdsourced digital platform designed to modernize public grievance redressal by addressing inefficiencies in traditional complaint resolution mechanisms. Citizens in India often face challenges due to poorly maintained infrastructure, including potholes, faulty streetlights, and uncollected garbage, with bureaucratic processes delaying resolutions. Infra-Alert provides a real-time, transparent, and automated system for lodging complaints, ensuring seamless submission with geolocation tracking and image uploads. Developed using HTML, CSS, JavaScript, PHP, and MySQL, Infra-Alert enables structured complaint routing, direct departmental complaints in PDF format, and Cloudinary-based cloud storage for issue images. A user-friendly dashboard allows citizens to track their complaint status, while authorities can efficiently view, solve, or reject issues, and improving accountability. The system streamlines data management, and enhances municipal responsiveness, bridging the gap between citizens and governing bodies. Future enhancements may include AI-driven complaint verification, automated prioritization, and chatbot-based assistance to further optimize urban issue resolution. This study underscores the necessity for digital transformation in public grievance systems, demonstrating how technology can enhance urban governance and service delivery.

Keywords--Public grievance redressal, Public Infrastructure Management, Geolocation Tracking, Reverse Geocoding, Cloud-based image storage, Real-time complaint tracking.

I. INTRODUCTION

Public infrastructure is a fundamental pillar of urban and rural development, directly impacting the safety, convenience, and well-being of citizens. However, unresolved civic issues such as potholes, broken streetlights, garbage accumulation, and faulty home electricity connections continue to disrupt daily life. Traditional complaint resolution mechanisms are often slow, bureaucratic, and lack transparency, leading to inefficiencies in addressing public grievances. Citizens face challenges in reporting issues due to cumbersome procedures, while authorities struggle with inefficient complaint routing, duplication of reports, and the absence of a centralized system for prioritizing and resolving complaints.

This paper presents Infra-Alert, a web-based crowdsourced platform designed to modernize public grievance redressal by automating complaint submission, tracking, and resolution. The system integrates Geolocation APIs for automated issue localization, Reverse Geocoding for precise address retrieval, Cloudinary for cloud-based image storage, and automated PDF generation for structured reporting. By providing a seamless, real-time, and transparent complaint management system, Infra-Alert enhances citizen engagement and enables authorities to efficiently track, verify, and resolve infrastructure issues.

Key contributions of this system include:

1. Automated location-based issue reporting using Geolocation APIs.
2. Efficient and transparent complaint management with real-time status tracking.
3. Cloud-based image storage for verification and documentation.
4. Automated PDF report generation for structured and streamlined issue processing.

By bridging the gap between citizens and municipal authorities, Infra-Alert transforms civic issue resolution from a reactive to a proactive approach. The system fosters government accountability, accelerates issue resolution, and ensures the effective maintenance of urban and rural infrastructure, ultimately improving public service efficiency and citizen satisfaction.

II. LITERATURE REVIEW

The inefficiencies in traditional grievance redressal systems have led to an increased demand for digital complaint reporting platforms in India. Existing solutions such as CPGRAMS [7] and the BBMP Sahaaya App [8] face limitations, including poor real-time tracking, lack of geolocation accuracy, and inadequate user engagement. Additionally, citizen-driven platforms like I Change My City [10] and Public Eye [11] attempt to provide a structured complaint management system but suffer from a lack of automation, real-time complaint tracking, and effective

departmental coordination. To address these challenges, Infra-Alert: A Crowdsourced Problem Reporting Platform has been developed as an innovative solution integrating cloud storage, geolocation APIs, and real-time tracking to enhance grievance redressal.

One of the primary challenges in grievance redressal systems is the lack of real-time tracking and transparency. CPGRAMS (Centralized Public Grievance Redress and Monitoring System) allows users to submit complaints related to public services, but it lacks real-time status updates and media integration [7]. Similarly, municipal-level platforms such as the BBMP Sahaaya App in Bengaluru, aimed at resolving complaints related to streetlights, potholes, and garbage collection, have proven ineffective [8]. Reports suggest that out of 12,000 complaints, only six were resolved in a month, highlighting inefficiencies in resolution mechanisms [9]. While I Change My City provides a user-driven approach for reporting civic issues, it lacks real-time tracking and automated status updates, resulting in delays in complaint resolution [10]. Research on online complaint management systems emphasizes the need for automated tracking and escalation mechanisms to ensure timely resolution. A study on grievance redressal systems proposes a blockchain-based solution for maintaining immutable records, ensuring transparency, security, and accountability [1]. While blockchain integration is an advanced approach, Infra-Alert focuses on practical enhancements using cloud storage and real-time dashboards to provide better complaint monitoring.

Another major concern is the ineffective geolocation mapping of complaints. The CPGRAMS Portal does not integrate real-time GPS tracking or reverse geocoding, making it difficult to locate and verify complaints efficiently [7]. Research on crowdsourced frameworks highlights that precise geolocation tracking can significantly improve issue resolution by allowing authorities to prioritize complaints based on location-specific needs [6]. The BBMP Sahaaya App suffers from location mis-mapping, as complaints are often logged at the user's current location rather than the actual issue location, leading to misrouted complaints and delays in action [9]. Similarly, Public Eye, an initiative to report civic issues like illegal parking and garbage dumping, lacks real-time geotagging and automated address extraction, making it difficult for authorities to pinpoint issue locations accurately [11]. Studies suggest that integrating Google's Reverse Geocoding API can significantly enhance the accuracy of complaint mapping [6]. Infra-Alert implements these features to ensure that complaints are correctly mapped to their respective locations, reducing response time and increasing operational efficiency.

A significant limitation of existing systems is the lack of crowdsourced verification and media evidence support. Current platforms primarily rely on manual complaint categorization and follow-ups, making the process slow and prone to inaccuracies. Studies have shown that media-based evidence submission, such as images, significantly improves complaint verification and resolution rates [3]. However, CPGRAMS and BBMP Sahaaya lack cloud-based storage for image uploads, limiting their effectiveness in verifying reported issues [8]. Similarly, I Change My City and Public Eye allow users to submit complaints but do not support scalable, cloud-integrated media storage, restricting authorities from efficiently accessing complaint-related

images for verification [10][11]. Infra-Alert integrates Cloudinary cloud storage for media uploads, ensuring secure storage and retrieval of complaint evidence. This feature allows authorities to verify complaints efficiently, improving response times and decision-making processes. Moreover, participatory planning and crowdsourcing have emerged as essential tools for smart city infrastructure management. Studies indicate that crowdsourcing enhances public engagement, accountability, and issue prioritization in urban governance [5]. An investigation into crowdsourcing for smart cities highlights that leveraging citizen-reported data can improve infrastructure maintenance, disaster management, and public safety [6]. Platforms like I Change My City and Public Eye encourage civic engagement but lack real-time feedback loops and automated departmental routing, leading to inefficiencies in issue resolution [10][11]. Infra-Alert incorporates direct user-authority interactions, enabling a transparent and participatory governance model. The limitations of existing grievance redressal systems necessitate the development of robust and cloud-integrated solutions. By leveraging real-time tracking, accurate geolocation mapping, crowdsourced verification, and participatory governance, Infra-Alert provides a scalable and efficient framework for urban complaint resolution.

III. PROBLEM STATEMENT

Efficient public infrastructure management is a fundamental challenge for urban authorities. Issues such as potholes, faulty streetlights, uncollected garbage, and home electricity failures significantly impact citizens' quality of life. However, traditional reporting mechanisms remain outdated and ineffective. Many municipalities still rely on manual complaint registration through phone calls, physical visits, or non-interactive online forms, which lack real-time tracking. This results in delayed responses, inefficient resolution processes, and limited citizen engagement. The existing infrastructure issue reporting systems face several critical challenges:

1. **Lack of Transparency and Complaint Tracking:** Citizens have no real-time visibility into the status of their complaints, leading to uncertainty about whether their concerns are being addressed. Authorities struggle to efficiently prioritize complaints based on severity and geographic location.
2. **Inefficient Image and Location-Based Reporting:** Many existing systems do not support image uploads, making issue verification difficult. The absence of GPS-based geolocation results in inaccurate problem identification, increasing the time required for resolution.
3. **Delayed and Unoptimized Resolution Process:** Manual logging and categorization of complaints slow down processing times. The lack of automated complaint routing forces authorities to filter complaints manually, delaying action on critical issues.
4. **Absence of Digital Documentation and Evidence:** Citizens often do not receive an acknowledgment or proof of complaint submission. Authorities lack structured historical data for analysis, planning, and decision-making.

This project aims to bridge the gap between citizens and urban authorities by developing a technology-driven, transparent, and efficient problem-reporting system. The Infra-Alert: Crowdsourced Problem Reporting Platform is designed to improve urban problem resolution, enhance civic engagement, and ensure a more responsive governance system. By integrating real-time issue tracking, automated complaint management, and structured documentation, the platform seeks to contribute to the development of smarter, safer, and well-managed cities.

IV. METHODOLOGY

The development of the Infra-Alert: Crowdsourced Problem Reporting Platform follows a systematic approach integrating web technologies, cloud storage, and geolocation services to ensure efficient and real-time issue reporting. The methodology is structured into multiple phases, as detailed below.

A. System Design

The platform was designed to address challenges in public infrastructure issue reporting, allowing citizens to report problems such as potholes, streetlights, garbage, and home electricity failures. The architecture facilitates seamless interaction between users, cloud storage, location services, and authorities.

System Components:

1. **User Interface:** Developed using HTML, CSS, and JavaScript to provide an interactive and responsive experience.
2. **Backend Server:** Implemented using PHP, responsible for form submissions, database interactions, and complaint routing.
3. **Database Management:** MySQL is utilized to store user details, complaint records, status updates, and location metadata.
4. **Cloud Image Storage:** Cloudinary is integrated for efficient complaint image storage and retrieval.
5. **Geolocation Services:** The Geolocation API extracts latitude and longitude, while Reverse Geocoding converts them into human-readable addresses.
6. **Authority Dashboard:** Facilitates complaint filtering, tracking, and resolution functionalities for municipal departments.

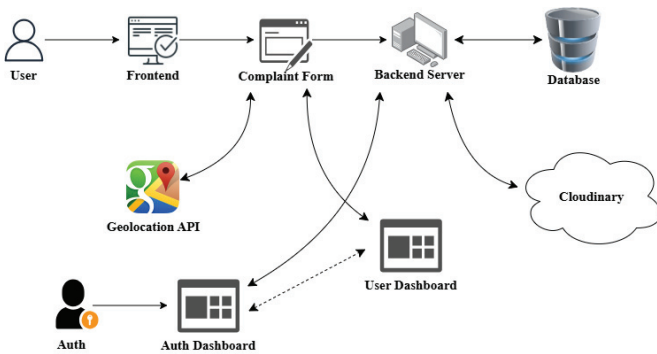


Fig. 1. System Architecture

The system architecture, as illustrated in Fig. 1, demonstrates the interaction between users, the frontend,

backend server, database, and external services. Users submit complaints via the frontend, which are processed by the backend and stored in the database. The platform integrates the Geolocation API for precise location tracking and Cloudinary for image storage. The user dashboard enables users to track complaint statuses, while the authority dashboard provides municipal departments with the necessary tools for efficient issue management and resolution.

B. Software Development

1) Frontend Development

The frontend is built using HTML, CSS, and JavaScript, ensuring a user-friendly interface with a responsive design. Key functionalities include:

- i. **Dynamic Complaint Submission Form:** Users can select issue categories and upload images.
- ii. **Google Maps Integration:** The platform fetches real-time location data using the Geolocation API.
- iii. **Interactive Dashboard:** Displays complaint status and updates for users.

2) Backend Development

The backend is developed using PHP, it handles:

- i. **Complaint Registration:** Validates and stores user submissions in the MySQL database.
- ii. **Image Handling:** Uploads and retrieves complaint images from Cloudinary.
- iii. **Status Updates:** Manages complaint progress and provides real-time updates.

3) Database Management

A MySQL database is structured to store essential data, including:

- i. **User Information:** Name, email, and authentication details.
- ii. **Complaint Records:** Issue category, image URL, submission time, and status.
- iii. **Geolocation Data:** Latitude, longitude, and formatted address for accurate mapping.
- iv. **Authority Logs:** Status updates, resolution timestamps, and resolved images.

4) Cloud Image Storage

Cloudinary is integrated for efficient image management, offering:

- i. **Image Compression & Optimization:** Enhancing load times.
- ii. **Secure Storage:** Preventing data loss.
- iii. **Dynamic Retrieval:** Allowing authorities to view complaint images for resolution purposes.

5) Geolocation and Reverse Geocoding Implementation

The Geolocation API extracts latitude and longitude for accurate issue reporting, while Reverse Geocoding ensures:

- i. Conversion of coordinates into human-readable addresses.
- ii. Accurate mapping of complaints to real-world locations.

- iii. Automated location fetching, reducing manual errors.

6) Authority Dashboard

The Authority Dashboard streamlines complaint management through:

- i. **Automated Complaint Categorization:** Ensuring reports reach the appropriate department.
- ii. **Status Management:** Authorities mark complaints as Pending, In Progress, Solved, or Rejected.
- iii. **Before/After Proof:** Uploading resolution images to verify problem resolution.

C. System Workflow

The system follows a structured workflow to streamline complaint registration, tracking, and resolution, as illustrated in Fig. 2, illustrates a streamlined public issue reporting process. Users submit complaints with relevant details and images, which are processed through the User Dashboard for status tracking and PDF generation. The Authority Dashboard manages complaints, updates statuses, and allows uploading resolved images, generating a Final Complaint Resolved PDF to ensure transparency and governance efficiency.

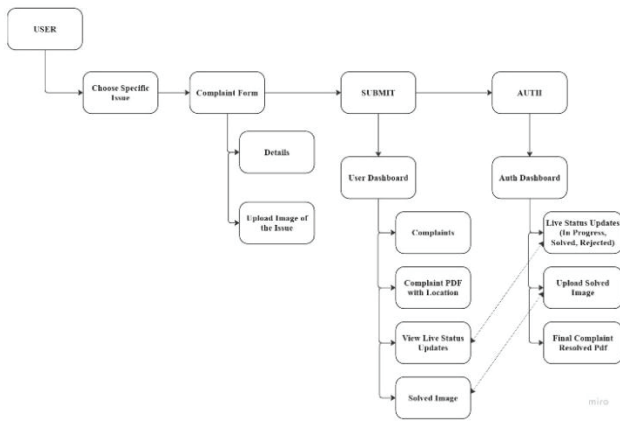


Fig. 2. Web Application Flow Chart

V. RESULTS AND DISCUSSION

A. Web application results

The main interface of Infra-Alert depicts in the Fig. 3, a citizen-driven platform for reporting public infrastructure issues. The homepage provides user-friendly navigation, enabling access to complaint registration, department details, and authentication for both users and authorities.

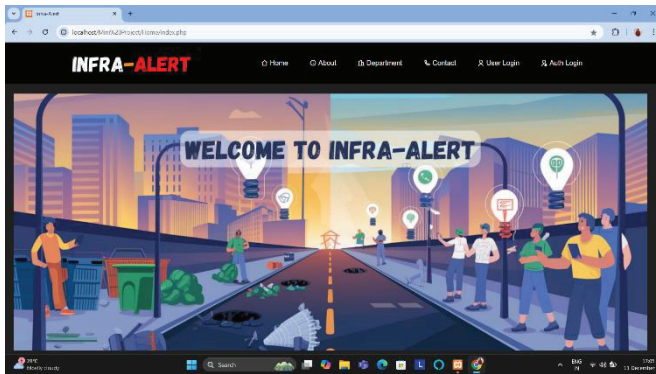


Fig. 3. Main Page

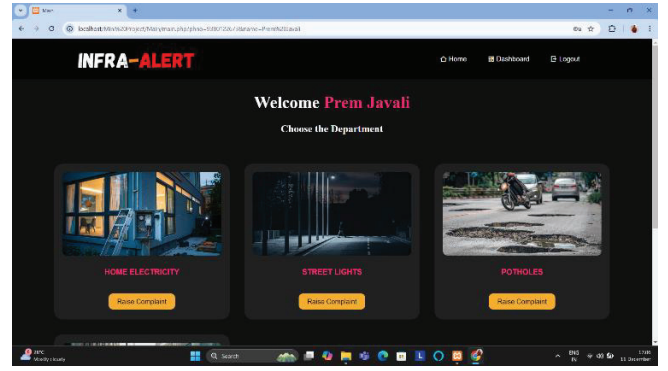


Fig. 4. Complaints Department Page

User complaint interface of Infra-Alert is depicted in Fig. 4, enabling citizens to report infrastructure issues by selecting relevant departments for streamlined resolution.

Fig. 5. Complaint Form

When a particular department is selected Complaint Form will be displayed as shown in Fig. 5, allowing users to submit issue details, select the category and department, and provide the issue image for efficient grievance redressal.

S No	C ID	Date	Phno	Category	Department	State	City	PDF	Status	Solved Image
1	C101	2024-12-03	9380122673	street_lights	Navi Mumbai Municipal Corporation	MH	Navi Mumbai		Pending	
2	C102	2024-12-03	9380122673	home_electricity	HUBLI Electricity Supply Company Limited (HESCOM)	KA	Belagavi		Solved	
3	C103	2024-12-04	9380122673	home_electricity	HUBLI Electricity Supply Company Limited (HESCOM)	KA	Belagavi		Solved	
4	C104	2024-12-04	9380122673	home_electricity	HUBLI Electricity Supply Company Limited (HESCOM)	KA	Belagavi		In Progress	
5	C105	2024-12-04	9380122673	home_electricity	HUBLI Electricity Supply Company Limited (HESCOM)	KA	Belagavi		In Progress	
6	C106	2024-12-04	9380122673	home_electricity	HUBLI Electricity Supply Company Limited (HESCOM)	KA	Belagavi		Pending	

Fig. 6. User Dashboard

Once the complaint is successfully submitted to a specific department user is redirected to the user dashboard as shown in Fig. 6, providing an overview of raised complaints, including their complaint ID, category, department, status, and resolution progress for effective tracking and management.

When authority logs into his account the Auth Dashboard is displayed as shown in the Fig. 7, enabling officials to track, verify, and update the status of raised complaints. It categorizes complaints into **in-progress**, **solved**, and **rejected** for efficient resolution management.

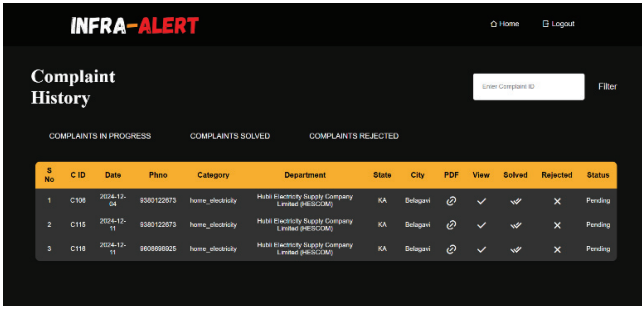


Fig. 7. Auth Dashboard



Fig. 8. Upload Solved Image

Once Auth solves the complaint raised by user needs to upload the solved image of the complaint as shown in the Fig. 8, it allows auth to upload the solved image and complaint is solved. Later, the Final pdf is generated with the complaint raised image as well as the solved image.

B. System Impact and Future Possibilities

1) Benefits:

The system enables real-time issue tracking, improving complaint resolution efficiency. Features like geolocation, cloud storage, and status tracking enhance user experience and transparency. Multi-department handling ensures better coordination among authorities.

2) Limitations:

The system relies on stable internet connectivity, which may affect real-time tracking in low-network areas. GPS accuracy can be compromised in enclosed or congested spaces, impacting location precision. Data security concerns arise with cloud storage, requiring strong encryption. Additionally, the system's effectiveness depends on active municipal participation; without it, issue resolution may be delayed despite advanced technical features.

3) Future Work:

Future enhancements can include AI-based issue categorization, predictive maintenance, and automated complaint routing. Integration with government databases and mobile payment options for municipal services can further streamline operations.

C. Comparison with Traditional technology

Table 1 presents the comparison of Infra-Alert and existing public grievance management platforms such as CPGRAMS (Government), I Change My City, and Public Eye. While traditional systems primarily facilitate basic complaint registration and tracking, they often lack the technological capabilities needed for efficient resolution. Infra-Alert, on the other hand, leverages advanced technologies such as cloud storage, geolocation APIs, and real-time tracking to enhance the grievance redressal process.

By integrating automation and real-time updates, Infra-Alert improves complaint visibility to users and authorities. Additionally, crowdsourced verification and media evidence storage enable authorities to validate complaints more effectively, reducing delays and miscommunication.

TABLE I. COMPARISON TABLE WITH EXISTING SYSTEM WITH INFRA-ALERT

Sl No.	Features	CPGRAMS [7]	I Change My City [10]	Public Eye [11]	Infra-Alert
1.	User Complaint Registration	✓	✓	✓	✓
2.	Real-Time Geolocation Tracking	×	×	×	✓
3.	Reverse Geocoding (Address Extraction)	×	×	×	✓
4.	Cloud Image Storage (Cloudinary)	×	×	×	✓
5.	Complaint Status Tracking	✓	✓	×	✓
6.	Authority Dashboard	✓	✓	×	✓
7.	Filter Complaints by Status	✓	✓	×	✓
8.	PDF Report Generation	×	×	×	✓
9.	Multiple Departments Handling	✓	×	×	✓
10.	Image Upload for Resolved Complaints	×	×	×	✓

Infra-Alert's advanced technology makes it a scalable and comprehensive solution for public grievance management.

- **Accuracy & Speed:** Real-time tracking ensures that exact locations are recorded, preventing delays in issue resolution.
- **Transparency & Accountability:** Cloud-based image verification and progress tracking build trust between authorities and the public.
- **Administrative Efficiency:** Multi-department handling and automated report generation minimize manual workload, allowing authorities to solve complaints effectively.

VI. CONCLUSION

The development of Infra-Alert: Crowdsourced Problem Reporting Platform, leveraging HTML, CSS, JavaScript, PHP along with Google Maps API and Cloudinary, presents a transformative approach to enhancing urban infrastructure issue reporting. By integrating real-time geolocation tracking, reverse geocoding, and cloud-based image storage, the system ensures accurate and efficient complaint registration. Features such as complaint status tracking, department-wise categorization, and automated PDF report generation enhance user experience and streamline issue resolution. The system's GPS-based location tagging and cloud integration provide seamless data management, improving transparency and responsiveness in handling

public grievances. The authority dashboard enables administrators to efficiently track, manage, and resolve complaints, ensuring accountability and structured decision-making. Real-time updates and automated complaint routing further enhance efficiency, reducing delays in addressing critical infrastructure concerns. This project successfully tackles inefficiencies in traditional complaint systems, such as delayed responses, lack of tracking, and mismanagement of reports, by providing a scalable, technology-driven solution. The modular and extensible architecture lays the foundation for future improvements, including AI-powered issue prioritization, predictive analytics for problem hotspots, and blockchain-based transparency mechanisms. Overall, Infra-Alert represents an innovative, data-driven, and scalable approach to public grievance redressal, contributing to a more responsive and citizen-centric urban governance system.

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